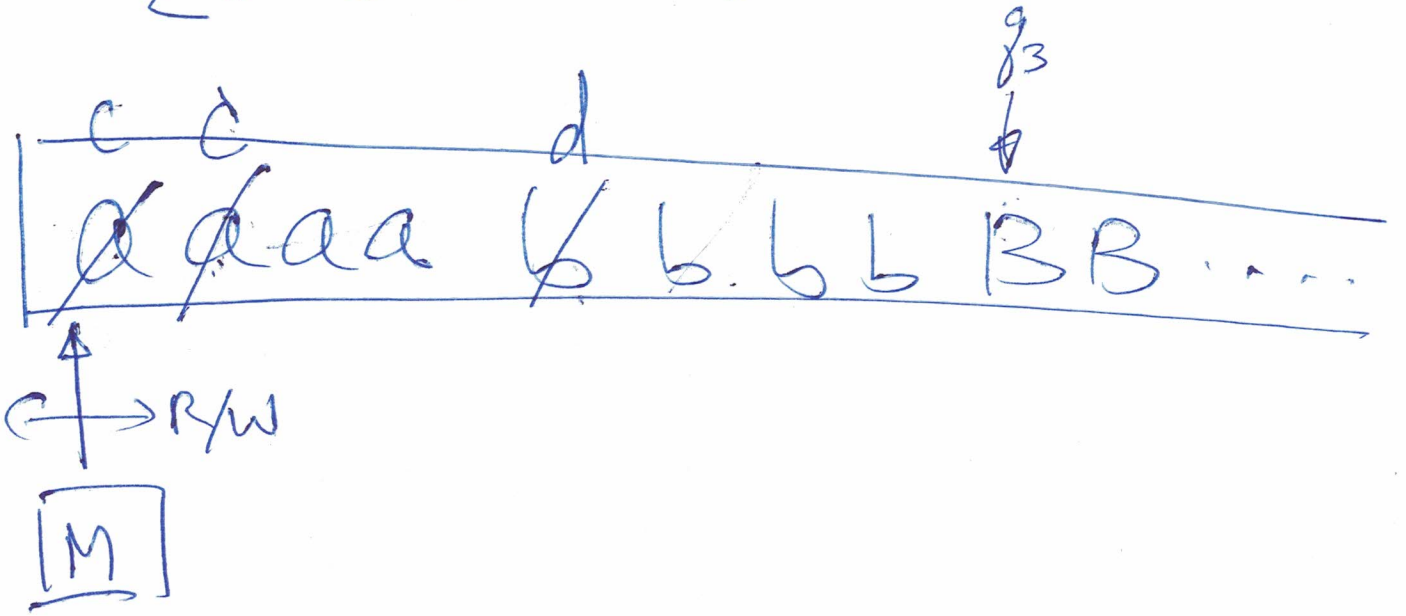
Cpts 317. 11/18.

Build a TM recognizing

$\{a^n b^n : n \geq 1\}$.



$\delta(q_0, a) = (q_1, c, R)$

$\delta(q_1, a) = (q_1, a, R)$

$\delta(q_1, b) = (q_2, d, R)$

$\delta(q_2, b) = (q_2, b, R)$

$\delta(q_2, B) = (q_3, B, S)$

$\delta(q_3, *) = (q_3, *, L)$, $* \neq c$,

$\delta(q_3, c) = (q_4, c, R)$

$\delta(q_4, a) = (q_5, c, R)$

$\delta(q_4, d) = (q_6, d, R)$ // no a found

$\delta(q_6, *) = (q_6, *, R)$ with $* \neq b, B$

$\delta(q_6, B) = (q_8, B, S)$ // q_8 is accepting.

$\delta(q_5, *) = (q_5, *, R)$ with $* \neq b, B$.

$\delta(q_5, b) = (q_3, d, S)$ // go back to
// the loop of q_3.

$\delta(q_5, B) = (q_7, B, L)$

$\delta(q_7, *) = (q_7, *, L)$ with $* \neq c, a$
// no a found

$\delta(q_7, c) = (q_8, c, S)$. // q_8 is accepting.

TMs can

1. compare two numbers,

2. do addition

$$a^n b^m c^{n+m}$$

3. can multiply

$$a^n b^m c^{n \cdot m}$$

⋮

So far:

reg languages	FA
c.f. languages	PDA
?	TM

When TM M runs on input word w, exactly one of the following happens:

1. M says yes on w
   (M enters an accepting state)
2. M crashes,
   (M says no on w)
3. M runs forever without saying yes/no.

C-programs and TMs are equivalent. When a program runs on an input,

1. it exits nicely, OR
2. it crashed befn it exits, OR
3. it runs forever.

TM M is an alg if M halts on all input.

(say yes or say no.)

L is accepted by a TM M if $L = L(M)$.

L is recognized by a TM M if $L = L(M)$ and M is an alg.

recursively enumerable

L is r.e. if L is accepted by a TM.

L is recursive if L is recognized by a TM.

Suppose that L is r.e. So, there is a TM M to accept it:

$\forall w$,

$w \in L \Rightarrow M$ says yes on w;

$w \notin L \Rightarrow M$ doesn't say yes on w.

Suppose that L is recursive. So, there is a TM M to recognize it:

an alg.

$\forall w$,

$w \in L \Rightarrow M$ says yes on w;

$w \notin L \Rightarrow M$ says no on w.

① Whether any recursive language is also r.e.?

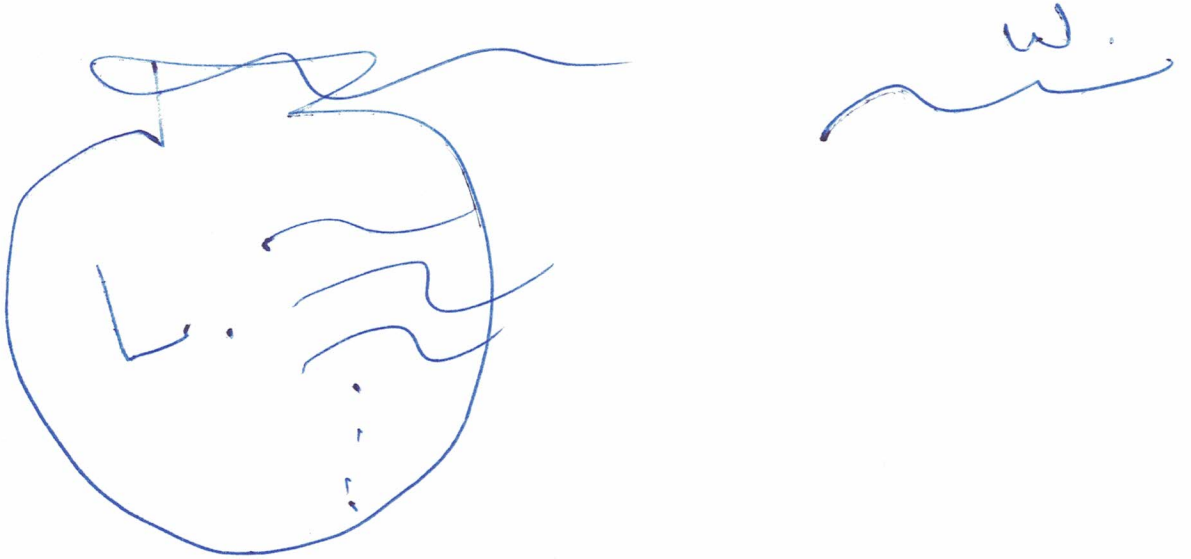
yes. obvious.

② Is there an r.e. language that is not recursive?

yes, proven by Turing.

(*) Given: a language L, and a word w. } problem

Question: $w \in L$?



When L is recursive, L is recognized by an alg M;

and this M is also an alg to solve (*).

w

When L is r.e., L is accepted by a TM M;

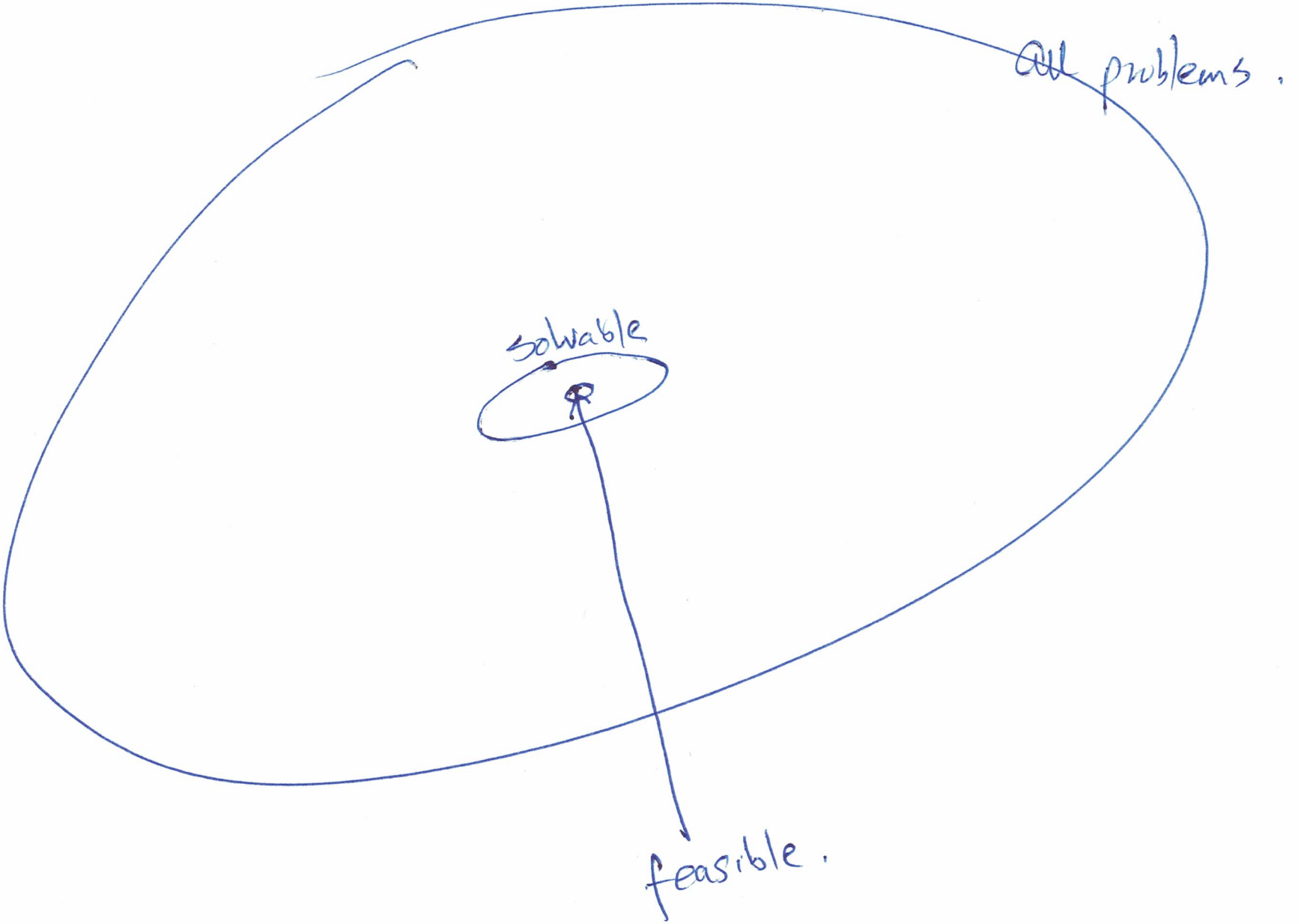
Then, $\forall w$,

$w \in L \Rightarrow M$ says yes on w;

$w \notin L \Rightarrow M$ says no on w, OR

M runs forever without saying yes.

Suppose that L is r.e., but not recursive. Then, we don't have ~~any~~ any alg to solve "$w \in L$?".


all problems.
solvable
feasible.

Proofs with TMs.

// Common mistake people make:

// We forget that in general, TM can run forever.

// pseudocode of TMs:

// ① { M_1 runs on w;
// if M_1 is happy, then M_2 runs on w.

// ② { M_1 runs on w;
// if M_1 is not happy, then M_2 runs on w.

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① is constructive. // Write a C-pgm to implement it.

② is not constructive.